MTH 201 Manipulating Power Series

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something, you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

1. Consider the integral $\int_0^2 e^{-x^3}\,dx$.

a) Find a series solution to the definite integral $\int_0^2 e^{-x^3}\,dx$.

b) How did you know that you could evaluate the series at $x = 2$ & $x = 0$?

2. Consider the value S given by $S = \frac{1}{e^0 \cdot 0!} + \frac{2}{e^1 \cdot 1!} + \frac{4}{e^2 \cdot 2!} + \ldots$. Determine $\ln(S)$.

3. Consider the function $f(x) = 2x\ln(1+x^2)$

a) Determine a power series for $f(x)$ about $x = 0$.

b) Based on the power series, what should the 13^{th} derivative of $f(x)$ be at $x = 0$?

c) Determine the exact value of the series $\sum_{n=0}^{\infty} \frac{(-1)^n}{2^{2n+2}(n+1)}$.

4. Determine the exact value of the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n \cdot 2^n}$.

5. Consider the function $f(x) = \frac{1}{(1+x)^2}$.

a) Use differentiation to help find a power series representation for this function. What is the radius of convergence for this power series?

b) Use your answer from part (a) to find a power series representation for $g(x) = \frac{1}{(1+x)^3}$ and state its radius of convergence.

c) Use your answer from part (b) to find a power series representation for $h(x)=\dfrac{x^2}{(1+x)^3}$ and state its radius of convergence.

d) Use your answer from part (a) to find a power series representation for $k(x)=\dfrac{1}{(1+2x)^2}$ and state its radius of convergence.

6. Find a function that is modeled by each of the following power series. Determine the radius of convergence for each of these power series.

a) $\displaystyle\sum_{n=0}^{\infty}\frac{(-1)^{n+1}x^{6n+4}}{(2n+1)!}$

b) $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n 3^{2n+1} x^{4n+4}}{(2n+1)}$

c) $\displaystyle\sum_{n=0}^{\infty} \frac{(-1)^n 5^{n+1} x^{n+2}}{(n+1)(n+2)}$

7. Consider the function $f(x)=\int_0^x \frac{\tan^{-1}(t)}{t}\,dt$. It can be shown that the discontinuity at $x=0$ is not a problem and the improper integral will converge.

a) Determine the Taylor series for the function $f(x)=\int_0^x \frac{\tan^{-1}(t)}{t}\,dt$.

b) Use your answer from part (a) to determine the value of $\int_0^1 \frac{\tan^{-1}(t)}{t}\,dt$ accurately to 2 decimal places.

8. Power series can be used to solve differential equations. Loosely speaking a differential equation is an equation that relate derivatives of a given function. Suppose that a country has at time t a population of $p(t)$, and that the rate of change of the population is given by $p'(t) = 2p(t)$. Here, the equation $p'(t) = 2p(t)$ is a differential equation. Solving this equation entails determining what function(s) $p(t)$ could satisfy this equation.

a) If we represent the total population of the country by a power series $p(t) = p(0) + p(1)t + p(2)t^2 + \ldots$, then use this power series and the differential equation to find a relationship between the coefficients of the power series. (*Hint: Think about how you can re-express the differential equation using the power series)*

b) Find the power series if $p(0) = 1$. *(Hint: Use your knowledge from part (a) to find the coefficientwise of the power series. These coefficients can be expressed in a way that involves powers of 2 and factorials.)*

c) Identify the power series as the Taylor expansion of some function $f(t)$.

9. Use Maclaurin series (not l'Hopital's rule) to find the following important limits.

a) $\lim_{x \to 0} \frac{\sin(x)}{x}$

b) $\lim_{x \to 0} \frac{1-\cos(x)}{x}$